Technical and Business Writing

Mid Long Questions (4 marks each) · Simple English, matched to the slides and notes

#	Long question	Source (slides / notes)	Priority
1	Writing Situation: Audience, Purpose, Context	L1: 5, 20–22, 25 · L2: 7–9 · Unit 1 §1.2 · Class	Must do
2	7 Cs of Communication	7 Cs notes, page 1–5	Must do
3	Email Writing	L1: 13–14 · Class	Must do
4	General vs Technical vs Business Writing	L1: 6–7, 11–12, 15–16 · Unit 1 §1.1	Backup
5	Technical Writing Process	L2: 4–5 · Unit 1 §1.5	Backup
6	Brainstorming (with mind map)	L2: 24–26, 28–29 · Class	Must do
7	Characteristics of Technical Writing	L1: 8, 23–24 · Unit 1 §1.3	Must do
8	Primary vs Secondary Sources + CRAAP Test	L2: 13–20 · Unit 2 §2.1, §2.5	Must do
9	Three Levels of Definition	Unit 3 §3.3 (notes only)	Backup
10	Designing a Questionnaire	Unit 2 §2.4 (notes only)	Backup

How to write a 4-mark long: Introduction (1–2 lines) → Table or points → One example → Conclusion (1 line).

LONG 1: Writing Situation (Audience, Purpose, Context)

Unit 1 (1.2) · Lecture 1 (slides 5, 20–21) · Lecture 2 (slide 7)

Q. Explain the writing situation: audience, purpose and context.

Introduction

Before writing, a writer must ask **three questions**: Who will read it? Why am I writing? Where will it be read? This is called the **writing situation**.

1. Audience: Who will read it?

Different readers know different things and need different things. There are 4 types:

Audience	What they want
Experts	Data and proof
Technicians	Clear steps
Managers	Short summary, cost and risks
Laypeople / Clients	Simple words and benefits

2. Purpose: Why am I writing?

What should the reader do after reading?

- **Inform:** tell what happened (e.g. progress report)
- **Instruct:** tell how to do something (e.g. installation guide)
- **Persuade:** make someone agree (e.g. proposal)

3. Context: Where and how will it be read?

On a phone, in a meeting or near a machine? Is it urgent? Context decides the **length, look and tone** of the document.

Example: same news, two readers

- **Developer:** "The login fails because the database table has no index."
- **Manager:** "Students cannot log in. It will take 2 days to fix and cost nothing."

Conclusion: When we know the audience, purpose and context, we can write the **right thing** in the **right way**.

Yaad karne ki trick: Kaun parhega? (Audience) · **Kyun** likh raha hoon? (Purpose) · **Kahan/kaise** parhega? (Context)

LONG 2: 7 Cs of Communication

7 Cs notes (Mind Tools)

Q. Explain the 7 Cs of communication with examples.

Introduction

The 7 Cs are a **checklist** for good communication. They make our emails, reports and meetings **clear and effective**.

C	Meaning	Example
1. Clear	Easy to understand; reader does not guess	Say which Daniel and what you want
2. Concise	Short; no extra words	Remove "kind of", "basically"
3. Concrete	Gives exact details	"Packs lunch in 1 minute"
4. Correct	No spelling or grammar mistakes	"two-**week**", not "two-**weak**"
5. Coherent	Logical; one topic in one email	Report and meeting in separate emails
6. Complete	Gives all needed information	Give the meeting **time and place**
7. Courteous	Polite and kind	"Could my team get extra time, please?"

Conclusion: If we follow the 7 Cs, people **understand** us easily and **trust** us.

Yaad karne ki trick: Saaf · Chhota · Pakka · Sahi · Jura · Poora · Tameez
(Clear · Concise · Concrete · Correct · Coherent · Complete · Courteous)

LONG 3: Email Writing

Lecture 1 (slides 13–14) · Class work · 7 Cs

Q. What is a professional email? Explain its parts and write an example.

Introduction

An email is the **most common way** to communicate at work. A good email is **short, clear and polite**.

Parts of a professional email

Part	What to write
1. Subject	What the email is about
2. Salutation	Polite greeting: "Dear Sarah,"
3. Purpose	Why you are writing
4. Details	Dates, times, names
5. Request	What you want, and **by when**
6. Closing	"Thank you," and your name

Bad email vs Good email (slide 14)

Bad email:
Subject: report
Hey, I need the project report ASAP. Send it over now.
Problems: rude, no deadline, no greeting, which report?

Good email:
Subject: Request: Updated Sprint 4 Progress Report
Dear Sarah,
Please send the updated Sprint 4 progress report by 3:00 PM today so we can review it before the client meeting.
Thank you,
Alex
Good points: clear subject, polite greeting, exact time, gives the reason.

Conclusion: A good email follows the **7 Cs**: it is clear, short, complete and polite.

Yaad karne ki trick: S-S-P-D-R-C: "**S**ubject, **S**alaam, **P**urpose, **D**etails, **R**equest, **C**losing"

If the paper says "Write an email to…": use these same 6 parts and only change the topic.

LONG 4: General vs Technical vs Business Writing

Lecture 1 (slides 6, 11, 15–16) · Unit 1 (1.1)

Q. Differentiate between general, technical and business writing.

Definitions

- **Technical writing:** explains technical information **clearly and correctly** to a **specific reader**.
- **Business writing:** writing used **at work** to share information and help **make decisions**.

General vs Technical writing

Basis	General Writing	Technical Writing
Purpose	To entertain	To inform or help do a task
Reader	Anyone	A specific reader
Tone	Emotional	Factual
Structure	Free	Headings, lists, tables
Words	Creative	Exact technical words

Technical vs Business writing

Basis	Technical Writing	Business Writing
Focus	Software and systems	Work and decisions
Documents	Manuals, API docs, SRS	Emails, memos, proposals
Reader	Developers, users	Managers, clients

Both: proposals and feasibility reports are technical **and** business documents.

Conclusion: A computer professional needs **both** technical and business writing.

Yaad karne ki trick: General = **dil** ki baat · Technical = **system** ki baat · Business = **faisle** ki baat

LONG 5: Technical Writing Process

Lecture 2 (slides 4–5) · Unit 1 (1.5)

Q. Explain the stages of the technical writing process.

Introduction

Good writing does not start by just typing. Writers follow **steps**.

Stage	What happens
1. Idea	Choose the topic or problem
2. Planning	Decide the reader, purpose and what to include
3. Gathering	Collect information
4. Reviewing	Check the information is correct and useful
5. Organizing	Put information in order (outline)
6. Drafting	Write the first version
7. Revising	Improve it and finish it

It is a cycle: the writer can **go back** to an earlier step if something is missing. It is not a straight line.

Example: while writing, the writer finds the cost data missing and goes back to collect it.

Conclusion: Following these steps gives a **clear and correct** document.

Yaad karne ki trick: "**I**dea **P**lan **G**ather **R**eview **O**rganize **D**raft **R**evise" → "**I**k **P**akistani **G**har **R**aat **O**r **D**in **R**oshan"

LONG 6: Brainstorming

Lecture 2 (slides 24–29) · Done in class

Q. What is brainstorming? Explain its techniques with an example.

Introduction

Brainstorming means **producing ideas** before deciding how they will appear in the final document. At this stage we do **not** write polished sentences.

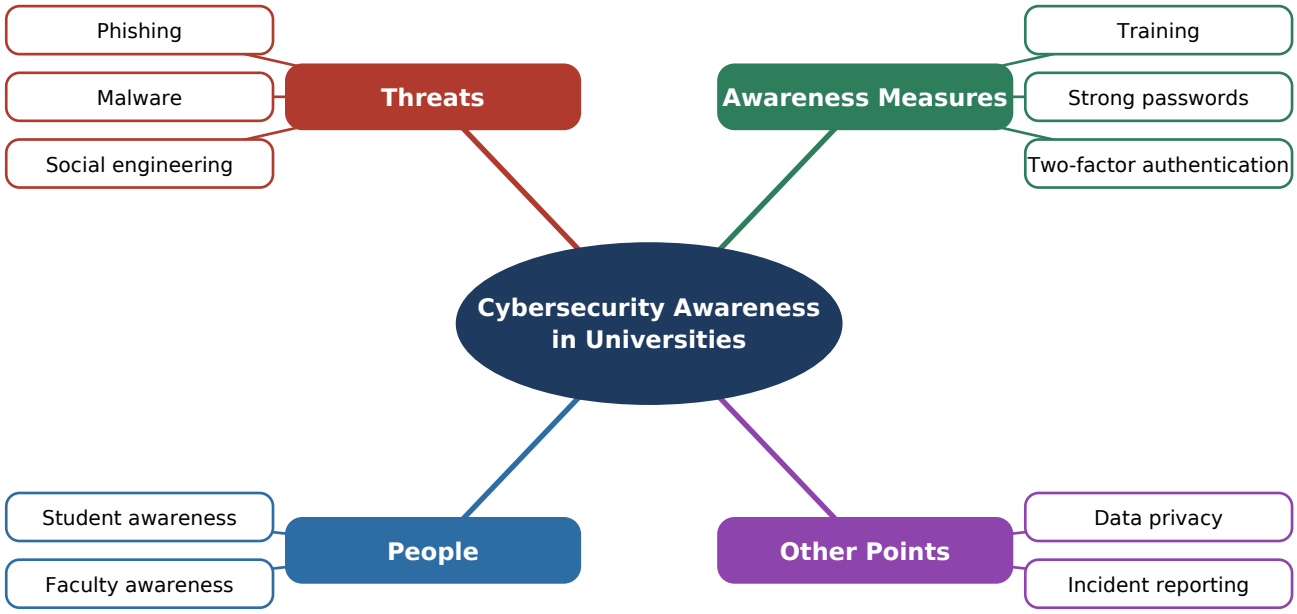
What we do in brainstorming

- **Generate ideas:** think of points and questions
- **Explore aspects:** look at the topic from different sides
- **Discover subtopics:** break a big topic into small parts
- **Delay editing:** do not reject an idea just because it is not well written

Techniques of brainstorming

Technique	Meaning	Example
1. Freewriting	Write ideas **non-stop** without editing	"Problems with our LMS…" and keep writing
2. Listing	**List everything** about the topic	security, cost, users, training
3. Questioning	Ask **Who? What? Why? When? Where? How?**	Questions show missing areas
4. Mind mapping	Put the **topic in the centre** and connect ideas around it	See the diagram below

Example: mind map (Cybersecurity Awareness in Universities)



From ideas to an outline



Formal outline (slide 29)

I. Introduction
II. Common Cybersecurity Threats
  A. Phishing B. Malware C. Social Engineering
III. Awareness Measures
  A. Training B. Strong Passwords C. Two-Factor Authentication
IV. Recommendations
V. Conclusion

Conclusion: Brainstorming helps the writer find **all the ideas first**, and then organize them into an outline.

Yaad karne ki trick: F-L-Q-M: "**F**ree **L**ist **Q**uestion **M**ap" (Freewriting, Listing, Questioning, Mind mapping)

In the paper: draw a small mind map (topic in the centre, 3–4 branches). It earns easy marks.

LONG 7: Characteristics of Technical Writing

Lecture 1 (slides 8, 23–24) · Unit 1 (1.3)

Q. Explain the characteristics of good technical writing.

Introduction

Good technical writing helps the reader **understand or do something** quickly and correctly.

Characteristic	Meaning
1. Clarity	Only one meaning; easy to understand on first reading
2. Accuracy	Facts, numbers and code are correct
3. Conciseness	Most information in fewest words
4. Logical Organization	Information in a clear order (headings, steps)
5. Objectivity	Facts, not feelings
6. Audience Awareness	Words and detail suit the reader
7. Precision	Exact numbers instead of vague words

Example (precision)

- **Vague:** "The software is slow when many users log in."
- **Precise:** "Response time goes from 120 ms to 4.2 s when users exceed 500 per second."

Myth: technical writing does **not** mean difficult words. "Use the application to complete the task" is better than "Utilization of the aforementioned application..."

Conclusion: These characteristics make a document **useful and trusted**.

Yaad karne ki trick: C-A-C-L-O-A-P: "**C**hai **A**ur **C**ake **L**ao, **O**r **A**ap **P**iyo"

LONG 8: Primary vs Secondary Sources + CRAAP Test

Lecture 2 (slides 13–20) · Unit 2 (2.1, 2.5)

Q. Differentiate primary and secondary sources. How do we evaluate a source?

Introduction

A document is only as good as its **information**. We get information from two kinds of sources.

Basis	Primary Source	Secondary Source
Meaning	Information you **collect yourself**	Information **others already collected**
Examples	Surveys, interviews, experiments, observation	Books, journals, reports, websites
Strength	Current and fits your problem	Quick to get; gives background
Weakness	Takes time	May be old or not fit your case

Evaluating a source: CRAAP test

Letter	Question
Currency	Is it **recent**?
Relevance	Does it **fit my topic**?
Authority	**Who** wrote it? Is the writer reliable?
Accuracy	Is there **evidence**? Can it be checked?
Purpose	**Why** was it written: to inform or to sell?

Example: for a current API feature, **official documentation** (updated recently) is better than an **anonymous blog** (no date, no evidence).

Conclusion: Good reports use **both** kinds of sources and check every source before using it.

Yaad karne ki trick: Primary = **khud** jama kiya · Secondary = **kisi aur** ka · CRAAP = **C-R-A-A-P**

LONG 9: Three Levels of Definition

Unit 3 (3.3) · notes only

Q. Explain the three levels of definition with examples.

Introduction

Technical fields have many terms. **Defining** them prevents misunderstanding.

Level	Meaning	Example
1. Informal	A quick meaning in **brackets**	"The firewall (a security system) blocked it."
2. Formal	**Term = Class + Features**	"A **router** is a networking device that forwards data packets."
3. Extended	A formal definition **explained more** with examples, history, comparison or parts	Cloud computing: examples (Gmail, AWS), parts (IaaS, PaaS, SaaS)

Common errors in formal definitions

- **Circular:** "Encryption is an algorithm that encrypts."
- **Class too broad:** "A compiler is a **thing** that..."
- **"Is when / is where":** "A deadlock **is when**..."

Conclusion: Choose the level of definition that **suits the reader**.

Yaad karne ki trick: Informal = **bracket** · Formal = **term + class + features** · Extended = **lambi wazahat**

LONG 10: Designing a Questionnaire

Unit 2 (2.4) · notes only

Q. How do we design and administer a questionnaire?

Introduction

A questionnaire collects the **same information from many people** quickly.

Types of questions

Type	Example
Closed (Yes/No)	"Do you use the LMS daily?"
Multiple-choice	"Which browser do you use most?"
Likert scale (1–5)	"The LMS is easy to use."
Ranking	"Rank these features."
Open-ended	"What one change would improve the LMS?"

Rules for good questions

- Avoid **leading** questions ("Don't you agree...?")
- Avoid **double-barrelled** questions ("Is it fast **and** easy?")
- Avoid **jargon** and **vague** words ("often")

Steps to administer

Short introduction → easy questions first → **pilot test** on 4–5 people → choose a fair sample → send, remind once, close on time.

Conclusion: Careful design gives **useful and reliable** answers.

Yaad karne ki trick: Types: "**C**hai **M**ein **L**aal **R**ang **O**cha" → Closed, Multiple, Likert, Ranking, Open